

Principles of Inheritance and Variation

Revision Notes – Quick Summary Cheat-Sheet

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Last-Minute Revision – Quick Summary

Key Dates & Discoverers

1856-63	Gregor Mendel - hybridisation experiments on garden pea
1865	Mendel publishes his work (unrecognised till 1900)
1900	de Vries, Correns & Von Tschermak - rediscover Mendel's work
1901	Karl Landsteiner - discovers ABO blood groups
1902	Sutton & Boveri - Chromosomal Theory of Inheritance
1903	Carl Correns - discovers incomplete dominance (<i>Mirabilis jalapa</i>)
1910s	T.H. Morgan - linkage & recombination (<i>Drosophila</i>)
1913	Alfred Sturtevant - first genetic map (map units/cM)
1924	Felix Bernstein - works out ABO inheritance pattern
1934	Folling - discovers Phenylketonuria (PKU)
1866	Langdon Down - first describes Down's syndrome

Memory Aids

Mendel's 3 Laws: "Dogs See Ice" → Dominance, Segregation, Independent Assortment

Deviations from Mendelism: "I Can't Make People Perfect" → Incomplete dominance, Co-dominance, Multiple alleles, Pleiotropy, Polygenic inheritance

Sex-linked disorders: "Happy Cats Hop" → Haemophilia & Colour blindness are X-linked recessive; Hypertrichosis is Y-linked

1. Genetics Basics & Mendel's Experiments

- Genetics = branch of biology studying inheritance & variation. Gene = Mendel's "factor"; allele = alternate form of a gene at a locus.
- Homozygous (TT/tt) vs Heterozygous (Tt); Monohybrid (1 character) vs Dihybrid (2 characters).
- Test cross = hybrid x homozygous recessive; Out cross = hybrid x homozygous dominant.
- Mendel (father of genetics) - garden pea (*Pisum sativum*), 1856-63, 7 pairs of contrasting characters.

2. Monohybrid Cross & Mendel's Laws

- TT x tt → F1 all Tt (tall) → selfing → F2 3:1 phenotypic, 1:2:1 genotypic.
- Law of Dominance: one factor of a dissimilar pair dominates the other. Law of Segregation: alleles separate cleanly during gamete formation.
- Punnett square - Reginald C. Punnett; F2 ratio derived from $(ax+by)^2$.
- Test cross case 1: TTxtt → all Tt (homozygous dominant). Case 2: Ttxtt → 1:1 Tt:tt (heterozygous).

3. Dihybrid Cross & Independent Assortment

- RRYY x rryy → F1 RrYy → selfing → F2 9:3:3:1 (9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green).
- F2 genotypic ratio 1:2:1:2:4:2:1:2:1.
- Law of Independent Assortment: two pairs of traits segregate independently of each other.
- Mendel's work published 1865, rediscovered 1900 by Hugo de Vries, Correns, Von Tschermak.

4. Deviations from Mendelism

- Incomplete dominance (Correns 1903): $RR \times rr \rightarrow Rr$ pink; F₂ 1:2:1 (Red:Pink:White).
- Co-dominance: both alleles express fully & independently; AB blood group, sickle cell carriers, roan cattle.
- Multiple alleles: ABO blood group - gene I with alleles I^A, I^B, i - 6 genotypes, 4 phenotypes.
- Pleiotropy: one gene $\rightarrow$ many traits; starch synthesis in pea, PKU, sickle cell anaemia.
- Polygenic inheritance: trait controlled by 3+ genes, additive effect; human skin colour.

5. Chromosomal Theory & Linkage

- Sutton & Boveri (1902): chromosome behaviour parallels gene behaviour.
- T.H. Morgan - *Drosophila melanogaster*; F₂ ratio deviated from 9:3:3:1 when genes were on the same (X) chromosome.
- Linkage = genes physically associated on a chromosome; Recombination = non-parental combinations.
- Sturtevant: recombination frequency $\rightarrow$ genetic maps. 1% crossing over = 1 map unit = 1 cM.

6. Sex Determination

- Henking - discovered "X body" (later, X chromosome). Allosomes = sex chromosomes; Autosomes = the rest.
- XO type (male heterogamety): female XX, male XO. Example - Grasshopper, Earthworm.
- XY type (male heterogamety): female XX, male XY. Example - Humans, *Drosophila*.
- ZW type (female heterogamety): female ZW, male ZZ. Example - Birds, some reptiles/fishes.
- Haplodiploidy (honey bee): fertilised egg (diploid, 32 chr) $\rightarrow$ female; unfertilised egg (haploid, 16 chr) $\rightarrow$ male.

7. Mutation & Pedigree Analysis

- Mutation = alteration of DNA sequence. Mutagens: physical (UV, gamma), chemical (nitrous acid), biological (viruses/bacteria).
- Point mutation = single base pair change (sickle cell anaemia). Frameshift mutation = insertion/deletion of a base pair.
- Pedigree analysis - traces a trait across a family tree using standard symbols (square=male, circle=female, filled=affected).

8. Genetic Disorders

- Autosomal dominant: mutant allele dominant; every generation affected. E.g. Huntington disease, Myotonic dystrophy.
- Autosomal recessive: carrier parents, consanguinity risk. E.g. Sickle cell anaemia (ch11), Thalassemia, PKU (ch12).
- X-linked dominant: Duchenne muscular dystrophy - no male-to-male transmission.
- X-linked recessive: Haemophilia A, Colour blindness.
- Y-linked (holandric): Hypertrichosis - father $\rightarrow$ all sons, never daughters.
- Chromosomal disorders: Down's syndrome (trisomy 21), Klinefelter's (47,XXY), Turner's (45,XO), Cri du chat (deletion, chromosome 5).